



Aror University of Art, Architecture, Design &
Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming Object Oriented Programming(Spring-2024)

LAB No. 2

Prepared by: Aurangzeb Magsi

Objective of Lab No. 1:

After performing lab1, students will be able to:

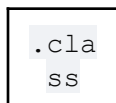
- Create and Run Hello World Program
- Use System.out.println() and System.out.print() for various outputs
- Use different variables and data types in java
- Use Arithmetic Operators in java
- Use Scanner class to take input from the user in java
- Use Comments in the Java Program
- Solve Real World Problems using all the above concepts

Recap of Last Lab:

In the Java ecosystem, **JDK**, **JRE**, and **JVM** represent three distinct layers that work together to create and run Java applications.

1. JVM (Java Virtual Machine)

The **JVM** is the core execution engine of Java. It is an abstract machine that provides a runtime environment for Java bytecode to execute.



- **Role:** It translates Java bytecode (files) into machine-specific code that your computer's CPU can understand.

- **Platform Independence:** While the JVM itself is platform-dependent (different versions exist for Windows, Mac, and Linux), it enables Java's "Write Once, Run Anywhere" (WORA) capability because the bytecode it runs is identical across all systems.
- **Key Tasks:** Loading code, verifying bytecode for security, executing code, and managing memory through **Garbage Collection**.



Aror University of Art, Architecture, Design & Heritage Sukkur.

2. JRE (Java Runtime Environment)

The **JRE** is a software package that provides everything necessary to **run** Java applications.

```
rt.jar
```

- **Components:** It contains the **JVM** plus the standard Java **class libraries** (like) and other supporting files.
- **Use Case:** You only need the JRE if you are an end-user who wants to run Java programs but does not intend to write or compile code.

```
JRE = JVM + Libraries + Supporting files
```

- **Formula:** .

3. JDK (Java Development Kit)

The **JDK** is a full software development kit used to **build** Java applications.

- **Components:** It is a superset of the JRE. It includes everything in the JRE (the runtime and JVM) plus **development tools**.

```
java
```

```
c
```

.jav
a

- **Tools:** Key tools include (the compiler that converts files

.cla
ss

jd
b

javado
c

to bytecode), (debugger), and (documentation generator).

- **Use Case:** Developers must install the JDK to write, compile, and test Java applications.

JDK = JRE + Development
Tools

- **Formula:** .

Objective-01: Create and Run First Java program (“Hello World”)

step 1: Write the following Code:

```
class HelloWorld{  
    public static void main(String [] args){  
  
        System.out.println("Hello this is first line");  
  
    } // end of the function  
  
} // end of the class
```

Step 2: save the file with .java extension in any drive/folder such as: HelloWorld.java



Aror University of Art, Architecture, Design &
Heritage Sukkur.

Step 3: open Command Prompt

Step 4: go to the desired Folder where you have saved your HelloWorld.java

Step 5: Set path as follows:

```
set path= [Paste your JDK's bin path here]
```

Step 6: Compile your code as follows: `javac HelloWorld.java`
if successfully runs, write the following Command
`java HelloWorld` [without `.java`]

Types of variables

In Java, there are three types of variables:

1. Local Variables
2. Instance Variables
3. Static Variables

1. Local Variables

Local Variables are a variable that are declared inside the body of a method.

2. Instance Variables

Instance variables are defined without the `STATIC` keyword. They are defined outside a method declaration. They are Object specific and are known as instance variables.

3. Static Variables

Static variables are initialized only once, at the start of the program execution. These variables should be initialized first, before the initialization of any instance variables.

Example: Types of Variables in Java

```
class Variables {  
    int data = 99; //instance variable  
    static int a = 1; //static variable  
    void method() {  
        int b = 90; //local variable
```



Aror University of Art, Architecture, Design &
Heritage Sukkur.

```
}  
}
```

Data Types in Java

Data types classify the different values to be stored in the variable. In java, there are two types of data types:

1. Primitive Data Types
2. Non-primitive Data Types

Primitive Data Types

Primitive Data Types are predefined and available within the Java language. Primitive values do not share state with other primitive values.

There are 8 primitive types: byte, short, int, long, char, float, double, and boolean

Integer data types

byte (1 byte)
short (2 bytes)
int (4 bytes)
long (8 bytes)

Floating Data Type

float (4 bytes)
double (8 bytes)

Textual Data Type

char (2 bytes)

Logical

boolean (1 bit) (true/false)



Aror University of Art, Architecture, Design &
Heritage Sukkur.

Input & Output

Output in Java Syntax:

```
System.out.println("Hello World");
```

Input in Java Syntax:

```
// import library
```

```
import java.util.Scanner;
```

```
// Creating scanner object
```

```
Scanner obj = new Scanner(System.in);
```

Taking input from user

```
int num = obj.nextInt(); // Integer Input
```

```
double num = obj.nextDouble(); // Double Input
```

Java Variable Type Conversion & Type Casting

A variable of one type can receive the value of another type. Here there are 2 cases. **Case 1)** Variable of smaller capacity is be assigned to another variable of bigger

```
double d ;  
int i = 10;  
d = i;
```

capacity.

This process is Automatic, and non-explicit is known as **Conversion**

Case 2) Variable of larger capacity is be assigned to another variable of smaller

```
double d = 10;  
int i;  
i = (int) d
```

capacity.

Type Cast
Operator



Aror University of Art, Architecture, Design &
Heritage Sukkur.

In such cases, you have to explicitly specify the **type cast operator**. This process is known as *Type Casting*.

In case, you do not specify a type cast operator; the compiler gives an error. Since this rule is enforced by the compiler, it makes the programmer aware that the conversion he is about to do may cause some loss in data and prevents **accidental losses**.

Example: To Understand Type Casting

```
class Demo {  
    public static void main(String args[]) {  
        byte x;  
        int a = 270;  
        double b = 128.128;  
        System.out.println("int converted to byte");  
        x = (byte) a;  
        System.out.println("a and x " + a + " " + x);  
        System.out.println("double converted to int");  
        a = (int) b;  
        System.out.println("b and a " + b + " " + a);  
        System.out.println("\ndouble converted to byte");  
        x = (byte)b;  
        System.out.println("b and x " + b + " " + x);  
    }  
}
```

Output:

```
int converted to byte  
a and x 270 14  
double converted to int  
b and a 128.128 128  
double converted to byte  
b and x 128.128 -128
```



Aror University of Art, Architecture, Design &
Heritage Sukkur.

Practice:

```
public class Task1{

    public static void main(String[] args) {

        System.out.println("== ////////////////////Student
Points\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\""); System.out.println();
        System.out.println("Name Lab Bonus Total");
        System.out.println("---- - - - - -");
        System.out.println("Joe \t43 \t7 \t50");
    }
}
```

Lab Exercise:

1. Create a java file called outputs.java, which outputs the data on console as shown below:

```
////////////////////////////////////////
==          Student Points          ==
////////////////////////////////////////

Name          Lab      Bonus      Total
----          - - - - -
Joe            43       7          50
William        50       8          58
Mary Sue       39       10         49
```

Practice:

```
public class Task1{
    public static void main(String[] args) {

        // 1. Addition of integers
        int a = 10;
        int b = 20;
        System.out.println("Addition of integers: " + (a + b));

        // 2. Addition of decimal numbers
        double x = 5.5;
        double y = 4.5;
        System.out.println("Addition of doubles: " + (x + y));
    }
}
```



Aror University of Art, Architecture, Design &
Heritage Sukkur.

```
// 3. String concatenation
String s1 = "Hello";
String s2 = "World";
System.out.println("String concatenation: " + s1 + " " + s2);

// 4. Number + String
int num = 100;
System.out.println("Number + String: " + num + " Java");

// 5. String + Number + Number
System.out.println("Result: " + 10 + 20);

// 6. Number + Number + String
System.out.println("Result: " + (10 + 20) + " Java"); }

}
```

Lab Exercise:

2. Observe the behavior of + operator in java and record the results of each of the statements shown below:

```
1 System.out.println ("This is a long string that is the " +
    "concatenation of two shorter strings.");

System.out.println ("The first computer was invented about" + 55 +
    "years ago.");

System.out.println ("8 plus 5 is " + 8 + 5);

System.out.println ("8 plus 5 is " + (8 + 5));

1 System.out.println (8 + 5 + " equals 8 plus 5.");
```

Practice:

```
public class Task1{

    public static void main(String[] args) {
```



Aror University of Art, Architecture, Design &
Heritage Sukkur.

```
String name = "Ali Khan";
int age = 17;
double gpa = 3.75; // Grade Point Average
char gender = 'M'; // Gender: 'M' for Male, 'F' for Female
boolean isForeigner = false; // true if foreigner, false otherwise
String studentID = "STU2025001"; // Unique student ID // Output the student data
System.out.println("Student Information:");
System.out.println("Name: " + name);
System.out.println("Age: " + age);
System.out.println("GPA: " + gpa);
System.out.println("Gender: " + gender); System.out.println("Foreigner: " + isForeigner);
System.out.println("Student ID: " + studentID);
    }
}
```

Lab Exercise:

3. You are creating a student database of at least five students for a school. Each student has a name, age, grade point average (GPA), Gender(M or F),foreigner(yes or No)and student ID. Create Variables for each of the attributes of a student. Value against each variable should be taken from the user as an input. Display all the results in a table format.

Practice:

```
public class Task1{

    public static void main(String[] args) {

        // Evaluate the expression (10 + 5) * (4 - 6) / 4
        int result = (10 + 5) * (4 - 6) / 4; // Display the result
        System.out.println("The result of the expression (10 + 5) * (4 - 6) / 4 is: " + result); }
}
```

Lab Exercise:

4. Write down a java program which displays the result of following mathematical expression



Aror University of Art, Architecture, Design &
Heritage Sukkur.

using System.out.println():

(10+5)*(4-6)/4

Practice:

```
public class Task1{

    public static void main(String[] args) {

        System.out.println("%t*t*t*t");
        System.out.println("%t*t*t*t");
        System.out.println("%t*t*t*t");
        System.out.println("%t*t*t*t");}

}
```

Lab Exercise:

5. Print out the following pattern using a single System.out.println(), with the help of escape sequences:

```
  _ _ _ _ _
 * * * * *
 *           *
 *           *
 *           *
 * * * * *
 * * * * *
```

Practice:

```
public class Task1{

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in); // 1 USD = 280.71 PKR (example rate, can be
updated)
        double exchangeRate = 280.71;
```



Aror University of Art, Architecture, Design &

Heritage Sukkur.

```
System.out.print("Enter the value in Dollars: ");  
double dollars = input.nextDouble();  
double rupees = dollars * exchangeRate;  
System.out.println("Value in Rupees: " + rupees);  
}
```



Lab Exercises:

6. Develop a Java program that calculates the volume of a cylinder. Prompt the user to input the radius and height, and then calculate and display the volume.
7. Write a Java program that converts a speed in miles per hour to kilometers per hour. Prompt the user to input a speed in miles per hour and then display the equivalent speed in kilometers per hour.
8. Develop a calculator program which takes, two numbers and an operation as input from the user and then performs the desired operations on both numbers and displays the output.